EDA Mini Project - A Coarse Study of Effect of Music on Mental Health

Introduction

Mental health has been an escalating public health issue in Hong Kong. According to Government publishment, the patients suffer from different degree of mental illness raised from 187,000 in 2011[i] to 305,700 in 2024[ii], which is 63% rise in only 13 years (See App1). On the other hand, there is research suggesting that music has certain effect to human mental health.[iii] While music is known to be part of everyday life,[iv] this Paper intends to study the relation between music and mental status from perspective of data analysis.

Study Approach and Scope

This Study follows scientific approach in analyzing the data between music and its effect on mental health.

Problem and Objective: Mental health issue is escalating in recent years. This Paper aims at looking for data to show music as a remediation to mental problem.

Hypothesis: There is relation between music and mental health.

Research and Scope: The study looks for recent research on music effect on mental health. Mental health is a general term while this research takes a deeper look at different types of mental disorder, including but not limited depression and anxiety as the most well-known illness. Secondly, analysis on music focuses not only on music listening, but also engagement like instruments and composition.

EDA and Data Cleansing: An exploratory data analysis (EDA) process is adopted to screen through the dataset applied in this Study. By EDA, the scope is defined with data to be focused. Irrelevant, erroneous and non-sense data and null value will be excluded and cleaned.

Analysis and Implication: The data is analyzed in univariate, bivariate and multivariate angles. Observation and implication are inspected upon the relation found in the data.

Limitation and Constraint: The Paper examines the limitation of the Study and constraint of the data, of which respective possible improvement is proposed. A conclusion is drawn with a prospect on future study.

Exploratory Data Analysis and Data Cleansing

It is an initial process to inspect the data set in order to understand the data value, categorize the structure and pattern, and identify the anomalies before further analysis and meaningful implication.

Dataset Description

i. Introduction to the dataset

The dataset is publicly available in Kaggle,[v5] which was a survey result collected via Google Form posted in various Reddit forums, Discord servers, and social media platforms in 2022. Posters and "business cards" were also utilized to advertise the form in libraries, parks, and other public locations. Respondents were not restricted by age or location. The dataset originally comprises 736 rows and 33 columns. After the data cleansing process described below, a dataset with 603 rows and 26 columns is utilized in this study.

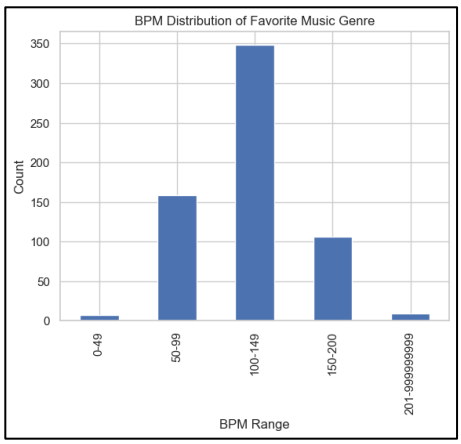
ii. Dataset characteristics

The dataset includes both quantitative and qualitative variables. Quantitative variables are continuous — age, hours of music listened daily, along with one discrete variable—BPM. Qualitative variables include nominal types like streaming service, music habits (e.g. instrumentalist, composer), favorite genre, as well as ordinal variables such as frequency of listening to 16 music genres, self-reported mental health scores (anxiety, depression, insomnia, OCD) and music effects

on mental health. The data uses float and object types, with values ranging across strings, integers, and booleans. (See App2) The columns of Hours per Day, Instrumentalist, Composer, Favorite Music Genre, BPM, Listening Frequency of 16 Music Genres, Anxiety, Depression, Insomnia, OCD, and Music Effects constitute the scope of this study.

Methodology of Data Cleansing – Data preprocessing & Rationale(s) behind

The data is semi-structured and may have some quality issues, so exploratory data analysis (EDA) and data cleaning are necessary. Potential data problems include:

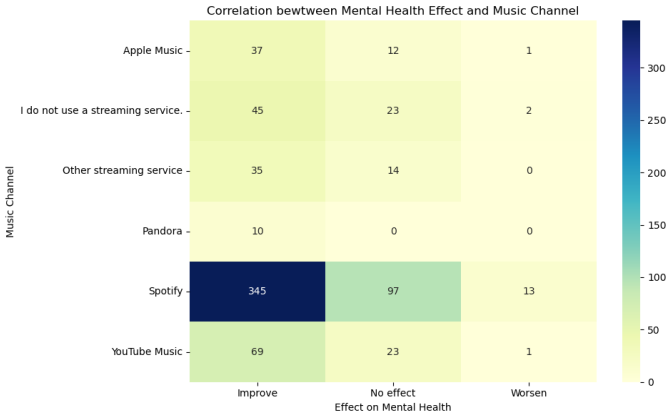


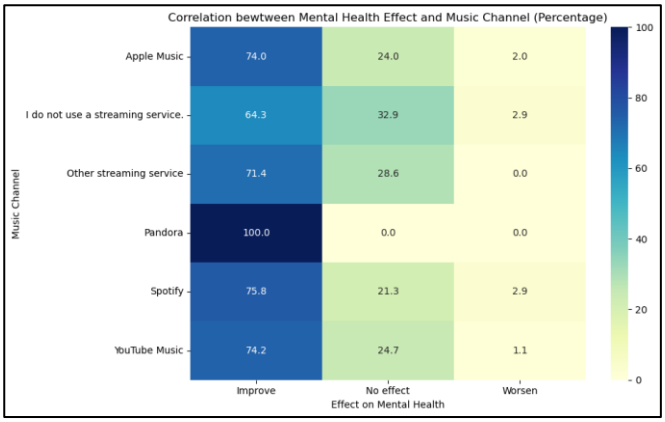
(1) Null values: Instrumentalist, Composer, BPM, and Music Effects contain null

(2) Erroneous or nonsensical inputs: Hours of music playing = 0, and BPM (Beats Per Minute) values outside the standard range. In general music literature, BPM (beats per minute) typically ranges from 50 to 200.[vi] By using Univariate Analysis, the right bar chart[vii] reveals that BPM values below 50 or above 200 are likely erroneous.

(3) Redundant or irrelevant data: Timestamp, Primary streaming service, and Permission that are not relevant to this study.

For primary streaming service, we conduct bivariate analysis with heatmap to study the correlation between the choice of music channel and its effect on mental health. A heatmap[viii](See App3) shows almost no correlation. The upper right heatmap[ix] looks deeper at each channel's effect based on usage counts, showing little relation except for Spotify—but it didn't account for overall channel popularity. The lower right heatmap[x] adjusts for this by showing the percentage effect per channel, confirming no significant difference. However, small sample sizes, like Pandora's 100% improvement with only 10 users, limit the conclusion. Considering all findings, the correlation is insignificant, so the Primary streaming service column was dropped from the dataset.





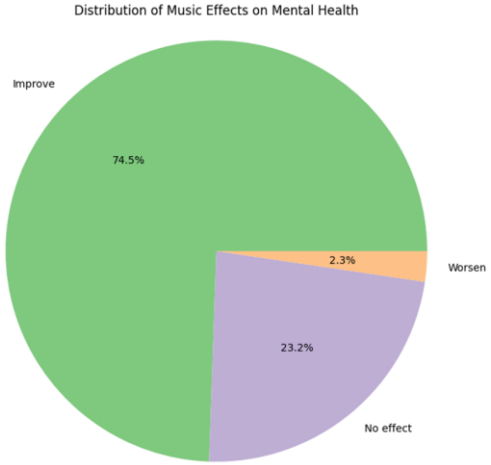
Analysis and Implication

i. Univariate Analysis

Direct View on the Survey Result on Mental Health Effect[xi]

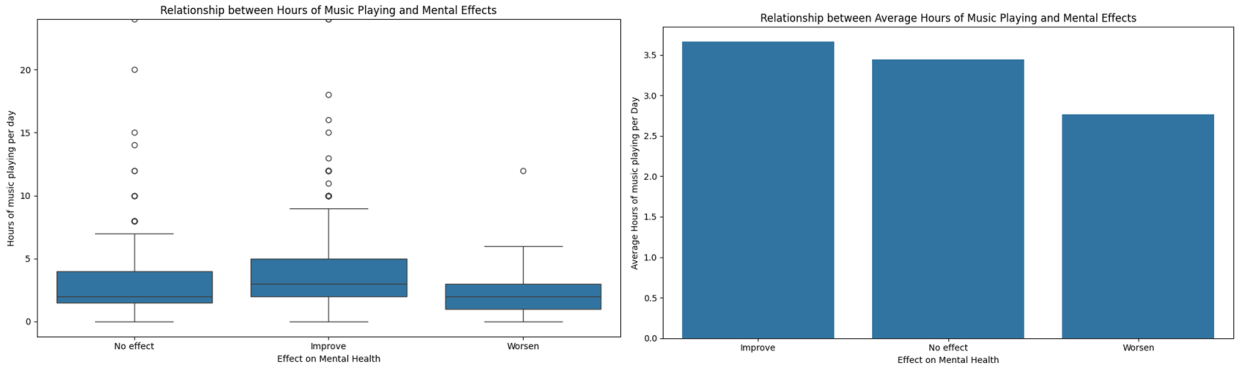
This is the first and basic analysis on music effect on mental status, generating a very crucial indication for the team to determine whether to continue the Study with the dataset.

It is found that over 75% of population opines that music (both listening and / or engagement (instrumentalist and / or composer)) has effect (either improve or worsen) to mental state. Among that, a majority reflected improvement in mental health. Meeting the hypothesis of the study, it worths deeper study on this survey and dataset.



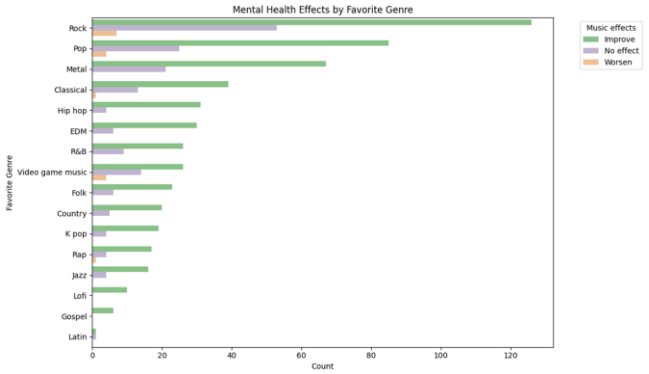
ii. Bivariate Analysis

Effect of Music Playing Duration on General Mental Health Improvement

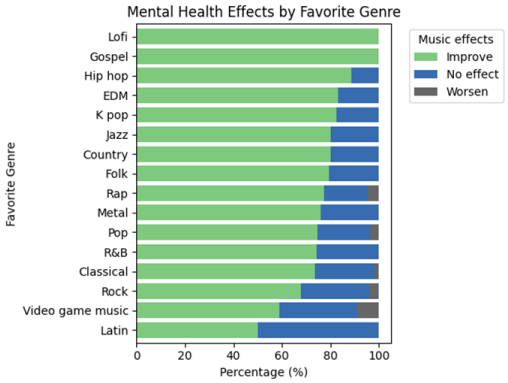


It is a bivariate analysis on the relation between the duration of music playing and its mental effect. The boxplot[xii] shows the distribution of hours/day of music playing of interviewees with different mental effect, whereas the bar chart[xiii] presents in terms of average hours of each mental effect. It is observed that candidates with improved mentality listened to music for around 3.5 hours/day in average. However, it does NOT mean it is a staggering or linear effect that listening less than 3.5 hours elicited no effect, NEITHER it reveals the causal relation that music playing less than 3 hours could worsen the health. It is found that the above relation is not obvious. Therefore, a wider analysis together with other attributes are carried out in multivariate view, see below section.

Relation of Favorite Music Genre and its Effect on General Mental State

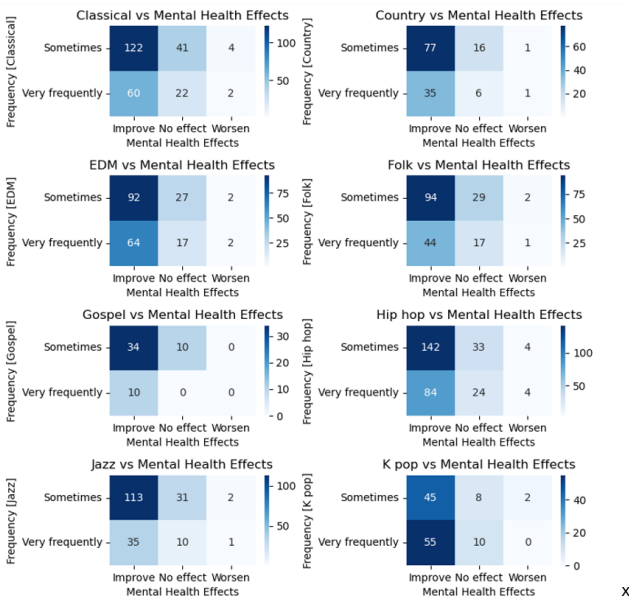


To drill in to "how" the relation between music and mental health builds, the perspective of genre is put under scope. The survey data collected the favourite music genre of the interviewees. It is used to plot two charts as shown. While the graphs attempt to highlight any specific genre that may bring more encouraging effect, note that the right bar chart[xiv] is based on the COUNT of the favourite genre. Take Rock as example which has the longest bar in improved mental health, it does NOT necessarily mean it is the most effective genre to bring healthier mental state. It may simply because Rock has the largest population of candidate choosing it as the favourite. Take a deeper look, it also has the largest count of 'No Effect' and even 'Worsen' result to the Mental Health.

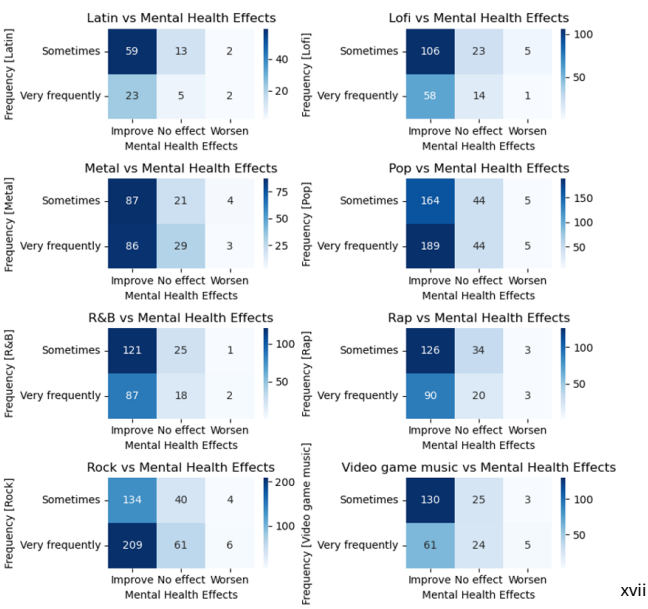


The left chart[xv] is therefore further investigating the effect by measuring its PERCENTAGE. It eliminates the illusion brought by the popularity of genre, trying to show the true effect of the genre to the mental health. Still, it has its defect and bias. Since it eliminates the consideration of population (i.e. count of genre chosen by candidate), it becomes the problem itself with the doubt on sampling size. For instances, Lofi and Gospel gave 100% improvement to mental health, whilst Latin brought the least (50%) improvement. It may NOT accurate to conclude Lofi and Gospel is the best genre while Latin is the most discouraging genre for mental health. Looking back the upper bar chart, it tells that those 3 genres have the least popularity and thus the sampling size as the limited representativeness for drawing a conclusion could be too assertive. In short, when analyzing the effect of genre to mental health, it is recommended to examine both COUNT and PERCENTAGE chart to prevent bias of popularity and sampling size.

Relation of Music Genre Listening Frequency and Mental State



xvi



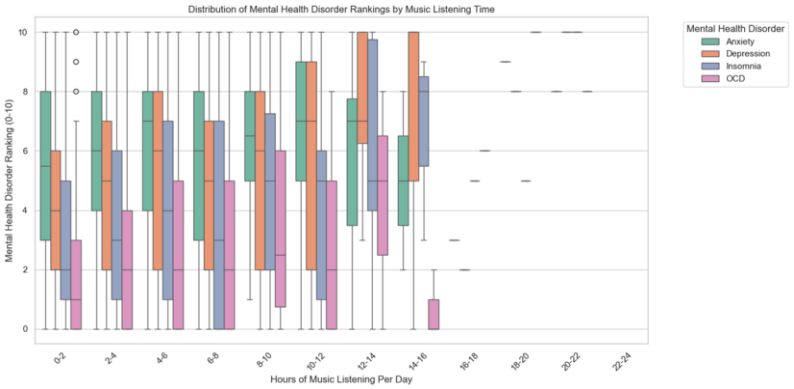
xvii

The heatmaps above illustrate the relationship between listening frequency for different music genres and their associated effects on mental health. It is important to note that the frequency categories "Never" and "Rarely" have been excluded from this analysis due to insufficient data points. Based on the heatmap analysis, Pop and Rock emerge as the most popular music genres among the 16 genres examined in the dataset. The majority of respondents report that listening to these genres improves their mental health, while only a small proportion indicate that these genres worsen their well-being. It should be noted that respondents may frequently listen to multiple music genres. The heatmaps depict the impact of listening to a specific genre frequently; however, this genre is not the sole factor influencing mental health. The analysis reveals that K-pop, Pop, and Metal demonstrate the most positive impact on mental health outcomes compared to other genres in the dataset. This suggests that these genres may have particularly beneficial effects on listeners' psychological well-being.

iii. Multivariate Analysis

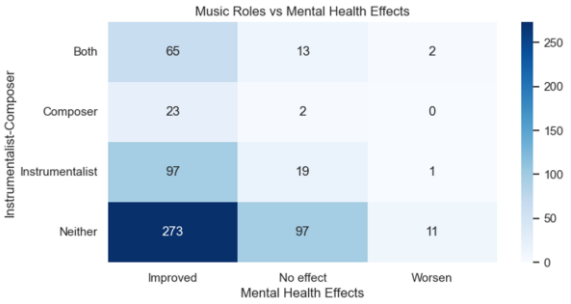
Effect of Music Playing Duration on Mental Health Disorder

The boxplots[xviii] (also see App4) show a positive correlation between daily music listening hours and mental health symptoms for Anxiety, Depression, Insomnia, and OCD. For Anxiety and Depression, symptom severity steadily rises from 6 to 14 hours of music listening, peaking in the 10–14 hour range. Insomnia symptoms also increase with more listening, with the highest levels in the 14–16 hour group. OCD symptoms rise gradually but remain lower and less consistent.



Overall, longer music listening (6–14 hours/day) is linked to more severe mental health symptoms, while lower listening (0–4 hours/day) corresponds to fewer symptoms. The trend for very high listening times (16+ hours) is unclear due to limited data. These results suggest people with more mental health struggles might use music as a coping tool, leading to longer listening times.

Effect of Music Engagement (Instruments / Composition) on Mental Health



The heatmap[xix] illustrates the correlation between different music roles and mental health effects. Music roles are defined by four categories: instrumentalist, composer, both (instrumentalist and composer), and neither (pure music listener). The majority of respondents across all music roles report that music improves their mental health, while only a small proportion report the opposite effect.

Music effects	Improve	No effect	Worsen	No effect/Improve	No effect/Worsen
Music role					
Both	0.812	0.162	0.025	0.200	6.5
Composer	0.920	0.080	0.000	0.087	-
Instrumentalist	0.829	0.162	0.009	0.196	19.0
Neither	0.717	0.255	0.029	0.355	8.818182

The table[xx] illustrates the proportion of music's effects on mental health across various musical roles. A smaller calculated ratio of "No Effect" to "Improve" and "No Effect" to "Worsen" indicates greater positive and negative impacts of music on mental health, respectively.

The analysis of the negative impact of music on mental health across musical roles is not included in this study due to an insufficient sample size. Notably, the ratio of music that benefits mental health to music that has no impact is highest among composers, followed by instrumentalists and individuals with dual musical roles, while it is lowest among pure music listeners. Given that the sample sizes for composers and instrumentalists are significantly smaller compared to that of pure music listeners, it is posited that active engagement in music, in addition to mere listening, may have a supplementary positive impact on mental health.

Challenges, Limitations and Possible Improvement

i. Challenges and Limitations

(1) Without a control group, it's hard to determine if mental health changes are directly related to music listening or other factors; (2) The data may be biased due to self-reporting inaccuracies in listening frequency and symptom rankings; (3) Difficulty in standardizing which reported listening hours are valid or not; (4) The lack of longitudinal data prevents tracking mental health changes over time; (5) There is no pre/post comparison to measure how music affects the 4 symptoms; and (6) The data does not allow for identification of which music genre most impacts mental health.

ii. Possible Improvements

(1) Utilize samples with 0 hours of music listening to serve as a baseline or control group for comparison. (2) Deeper analysis to list out outliers for consideration of exclusion. (3) To research on average hours of playing, define the normal range in consideration of 25 to 75 percentile. (4) Seek clarification from the survey inspector or designer to better understand unclear data points. (5) Expand multivariate analysis, e.g. Fav genre X Frequency of genre X Effect.

Conclusion

To conclude, listening to music generally benefits mental health to a certain extent, while other forms of music engagement—such as playing instruments and music composition—may provide supplementary positive effects on well-being. For a deeper investigation based on the results of this study, it is essential to reexamine and refine the research and survey methods. Some topics worth exploring further include: the correlational relationship between music and mental health, as opposed to the causal relationship assumed in this study; the impacts of music engagement beyond listening; the extent to which music can affect mental health in relation to other possible factors; and the specific music genres that have the most pronounced impact on mental health. Ultimately, this study highlights the importance of music as a potential tool for enhancing mental health and encourages further research in this area to deepen our understanding of its complexities. By exploring these topics, future research can contribute to developing effective interventions and strategies that harness the benefits of music for mental well-being.

<END>

Appendix

App1: Number patients suffering from Mental Illness (2011-2024)

Year	Patients with Mental Illness
2011 – 2012	187,000
2015 – 2016	220,000
2019 – 2020	270,700
2023 – 2024	305,700

App2: Dataset characteristics

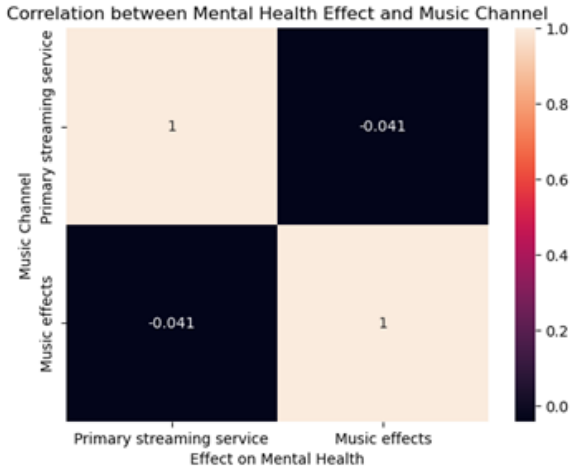
No.	Column (Variables)	Description	Data Type (Pandas/Numpy)	Variable Categories	Variable Type	Variable Data Type (general)
1	Timestamp	Form submission date and time	object	Qualitative	Nominal	String (Datetime)
2	Age	Respondent's age	float	Quantitative	Continuous	Integer
3	Primary streaming service	Respondent's primary streaming service	object	Qualitative	Nominal	String
4	Hours per day	Daily music listening hours	float	Quantitative	Continuous	Integer
5	While working	Does the respondent listen to music while studying/working? (Yes/ No)	object	Qualitative	Nominal	Boolean
6	Instrumentalist	Does the respondent play an instrument regularly? (Yes/ No)	object	Qualitative	Nominal	Boolean
7	Composer	Does the respondent compose music?	object	Qualitative	Nominal	Boolean
8	Fav genre'	Respondent's favorite or top genre	object	Qualitative	Nominal	String
9	Exploratory	Does the respondent actively explore new artists/genres? (Yes/ No)	object	Qualitative	Nominal	Boolean
10	Foreign languages	Does the respondent regularly listen to music with non-fluent lyrics? (Yes/ No)	object	Qualitative	Nominal	Boolean
11	BPM	Beats per minute of favorite genre	float	Quantitative	Discrete	Integer
12-27	*Frequency [Music Genre]	Frequency of the respondent's listening to a music genre (Never, Rarely, Sometimes, Very frequently)	object	Qualitative	Ordinal	String
28	Anxiety	Self-reported anxiety (0-10)	float	Qualitative	Ordinal	Integer
29	Depression	Self-reported depression (0-10)	float	Qualitative	Ordinal	Integer
30	Insomnia	Self-reported insomnia (0-10)	float	Qualitative	Ordinal	Integer

No.	Column (Variables)	Description	Data Type (Pandas/Numpy)	Variable Categories	Variable Type	Variable Data Type (general)
31	OCD	Self-reported OCD (0-10)	float	Qualitative	Ordinal	Integer
32	Music effects	Does music affect the respondent's mental health? (No effect / Improve / Worsen)	object	Qualitative	Ordinal	String
33	Permissions	Permissions to publicize data	object	Qualitative	Nominal	String

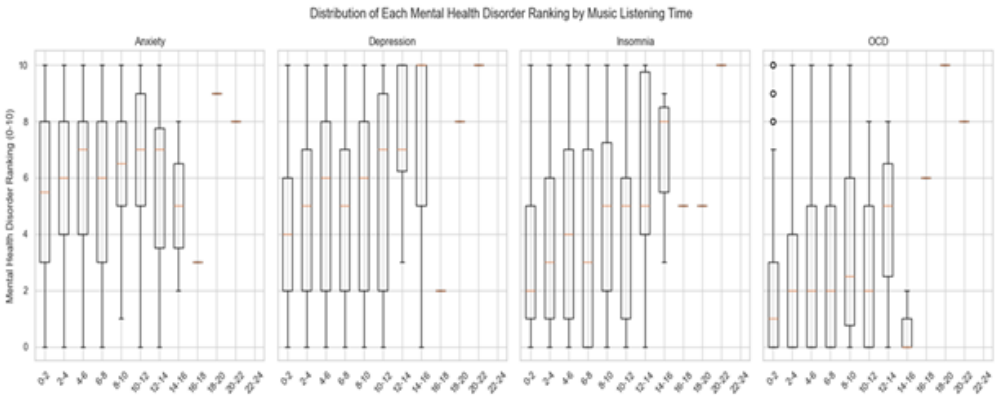
* The music genres are Classical, Country, EDM, Folk, Gospel, Hip Hop, Jazz, K-Pop, Latin, Lo-Fi, Metal, Pop, R&B, Rap, Rock, and Video Game Music.

Data in green: columns adopted in the scope of this Study

App3: Heatmap of Correlation between Mental Health Effect and Music Channel



App4: Boxplot of Distribution of Each Mental Health Disorder Rankings by Music Listening Time



Reference

[i] Food and Health Bureau, Mental Health Review Report (2018) para1.4.1 <https://www.healthbureau.gov.hk/download/press_and_publications/otherinfo/180500_mhr/e_mhr_full_report.pdf> (Last accessed on 2 Aug 2025)

[ii] Legislative Council Secretariat, Heath status of Hong Kong citizens (ISSH11/2024) <https://app7.legco.gov.hk/rpdb/en/uploads/2024/ISSH/ISSH11_2024_20240531_en.pdf> (Last accessed on 2 Aug 2025)

[iii] Angela Chen, 'Music, Mental Health, and Mood Regulation: A Data-Driven Approach to Understanding the Role of Music in Emotional Well-being' (2024) 2(4) American Journal of Student Research

[iv] Emily Carlson, Johanna Wilson, Margarida Baltazar, Deniz Duman et al., 'The Role of Music in Everyday Life During the First Wave of Coronavirus Pandemic: A Mixed-Method Exploratory Method' (2021)12:647756 Frontiers in Psychology 1

[v] Catherine Rasgaitis, 'Music & Mental Health Survey Results'. (2022, November 21). Kaggle. <https://www.kaggle.com/datasets/catherinerasgaitis/mxmh-survey-results> (Last accessed on 2 Aug 2025)

[vi] Rebecca Carson, 'Tempo Indications Reference Sheet', Hoffman Academy <https://www.hoffmanacademy.com/blog/tempo-indications-reference-sheet> (last accessed on 2 Aug 2025)

Supplement: Title of Figure

[vii] Bar Chart of BPM Distribution of Favorite Music Genre

[viii] Heatmap of Correlation between Mental Health Effect and Music Channel

[ix] Heatmap of Correlation between Mental Health Effect and each Music Channel (Count)

[x] Heatmap of Correlation between Mental Health Effect and each Music Channel (Percentage)

[xi] Pie Chart of Distribution of Music Effects on Mental Health

[xii] Boxplot of Relationship between Hours of Music Playing and Mental Effects

[xiii] Bar Chart of Relationship between Average Hours of Music Playing and Mental Effects

[xiv] Bar Chart of Mental Health Effects by Favorite Genre

[xv] Stacked Bar Chart of Mental Health Effects by Favorite Genre

[xvi] Heatmap Analysis of Listening Frequency to Music Genres and Their Impact on Mental Health-1

[xvii] Heatmap Analysis of Listening Frequency to Music Genres and Their Impact on Mental Health-2

[xviii] Boxplots of Distribution of Each Mental Health Disorder Rankings by Music Listening Time

[xix] Heatmap Representation of Music Roles and Their Impact on Mental Health

[xx] A Comparative Analysis of Different music Roles and the Impact of Music on Mental Health